

Tutorial sheet - (1)

① Which of the following sequences $\{a_n\}$ converge, and which diverge? find the limit of each convergent sequence.

i) $a_n = \frac{n^r}{e^n}$

ii) $a_n = \frac{1-2n}{1+2n}$

iii) $a_n = \frac{n+3}{n^r+5n+6}$

iv) $a_n = 1 + (-1)^n$

v) $a_n = \frac{n^r-2n+1}{n-1}$

vi) $a_n = \frac{n}{2^n}$

vii) $a_n = \frac{8^n n^n}{2^n}$

viii) $a_n = \frac{\log n}{n^n}$

ix) $a_n = \frac{n!}{n^n}$

x) $a_n = \frac{5^n}{7^n}$

xi) $a_n = \frac{\log n}{n}$

xii) $a_n = \left(\frac{n+1}{n-1}\right)^n$

② prove that the sequence $\{a_n\}$ is defined by

$$a_n = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{n!} \text{ is convergent.}$$

③ show that the sequence $\{a_n\}$ defined by $a_n = \left(1 + \frac{1}{n}\right)^n$ is convergent.

④ show that the sequence $\{a_n\}$ defined by $a_1 = \sqrt{2}$ and

~~4~~ $a_{n+1} = \sqrt{2a_n}$ converges to 2.

⑤ Show that the sequence $\{a_n\}$ where $a_1 = 1$ and

~~$a_n = \sqrt{2 + a_{n-1}}$~~ $a_n = \sqrt{2 + a_{n-1}}$ for $n \geq 2$ is convergent and converges to 2.

⑥ Show that the sequence $\langle a_n \rangle$ where $a_n = \frac{1}{(n+1)^2} + \frac{1}{(n+2)^2} + \dots + \frac{1}{(n+n)^2}$ converges to zero.

⑦ Prove that the sequence $\langle a_n \rangle$ where $a_n = \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n}$ is convergent and its limit lies between $\frac{1}{2}$ and 1.

⑧ Prove that the sequence $\{a_n\}$ defined by $a_1 = \sqrt{7}$, $a_{n+1} = \sqrt{7 + a_n}$ converges to the positive root of the eqn $x^2 - x - 7 = 0$

⑨ If a_1, b_1 are two positive unequal numbers and a_n, b_n are defined as $a_n = \frac{1}{2}(a_{n-1} + b_{n-1})$, $b_n = \sqrt{a_{n-1}b_{n-1}}$ for $n \geq 2$, prove that two sequences $\{a_n\}, \{b_n\}$ are monotonic, one increasing and the other decreasing and that they tend to the same limit.

⑩ If $a_1, b_1 > 0$ and $a_n = \sqrt{a_{n-1}b_{n-1}}$ and $b_n = \frac{2a_{n-1}b_{n-1}}{a_{n-1} + b_{n-1}}$, prove that (i) $\{a_n\}$ and $\{b_n\}$ are monotonic (ii) $\{a_n\}$ and $\{b_n\}$ both converge to a common limit.